

Session Objectives

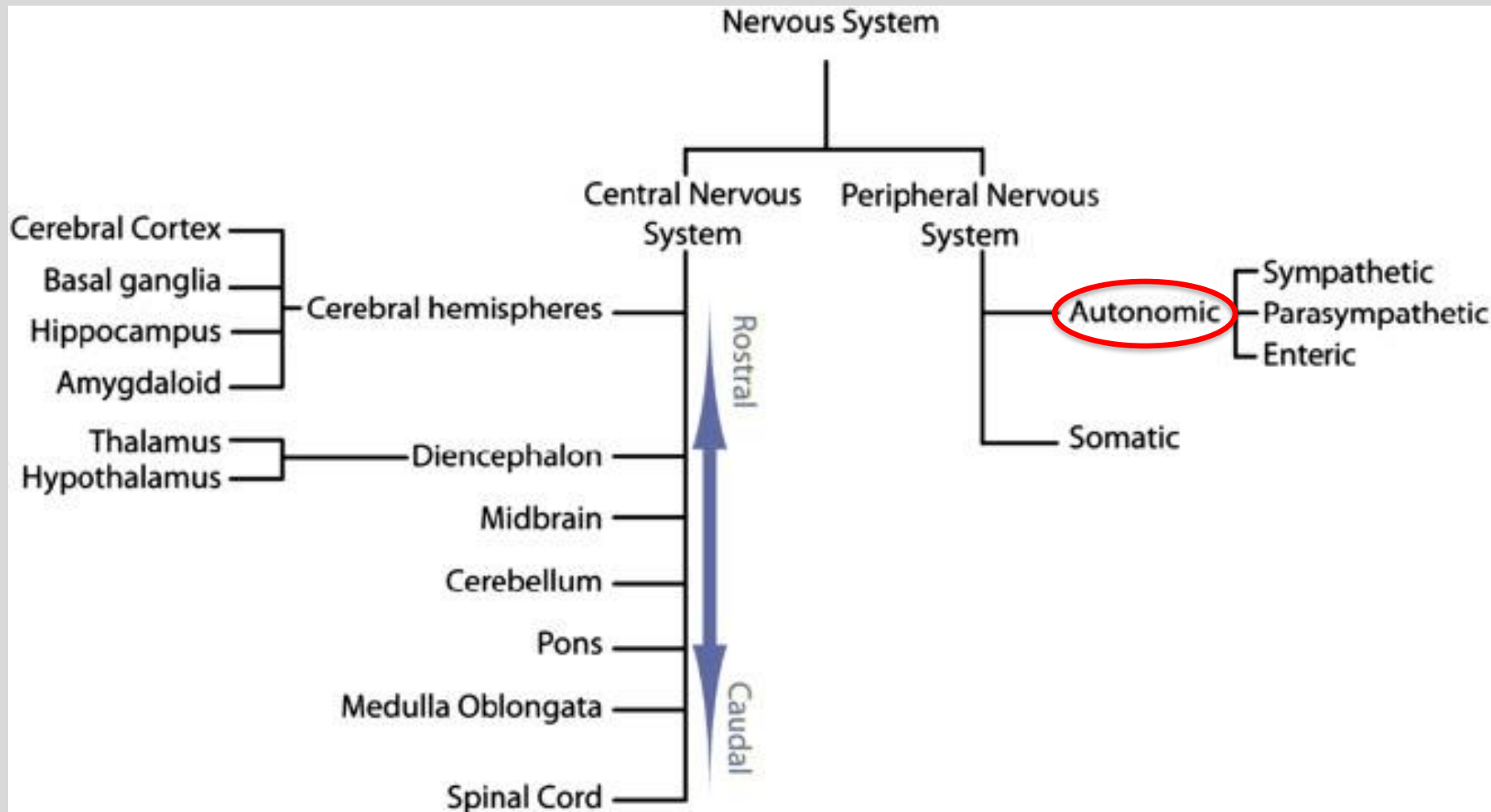
At the end of this lecture, students will be able to:

1. Describe the anatomy and neurotransmitters of the autonomic nervous system (ANS)
2. Depict the actions of the sympathetic and parasympathetic nervous system
3. Depict anatomy and physiology of the enteric nervous system and its neuromodulation
4. Comprehend the physiological basis of autonomic reflexes and several autonomic disorders

Organization of the nervous system

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

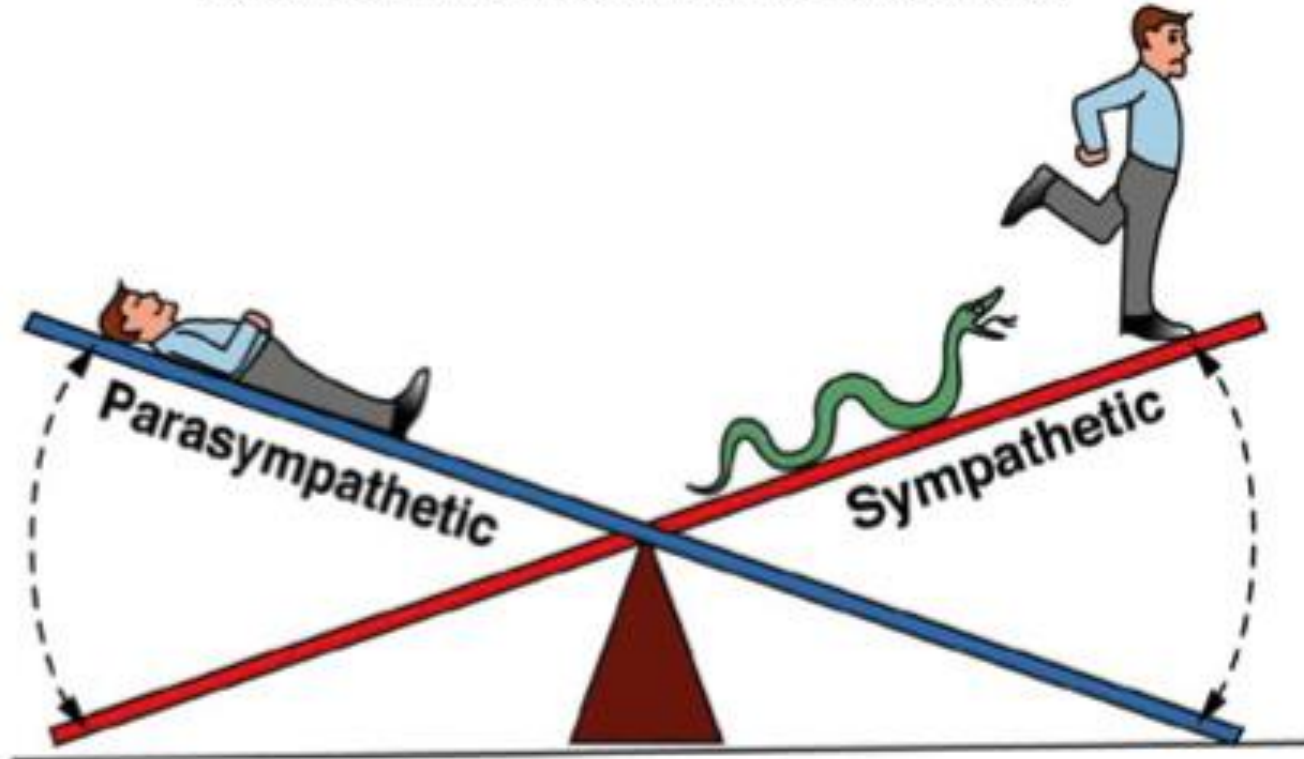


ANS and Homeostasis

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

**Homeostasis is a Synergistic Balance
between the Autonomic Branches**



'Rest, Heal & Digest':
Parasympathetic
activity dominates.

'Fight or Flight':
Sympathetic
activity dominates.

<https://www.replenishyourlife.co.uk/post/the-role-of-the-autonomic-nervous-system-in-cfsme>

Autonomic Innervation of Selected Body Structures

Structure	Sympathetic	Parasympathetic
eye - pupil (iris):	dilation	constriction
heart:	increased rate & force	decreased rate
lung bronchioles:	dilation	constriction
gut wall:	decreased motility	increased motility
gut sphincter:	constrict	relax
bladder detrusor:	relax	contraction
bladder-urethra smooth sphincter:	constrict	relax
penis etc.:	ejaculation	erection
gallbladder bile & duct:	relax	contract
salivary glands:	concentrated viscous saliva	abundant watery saliva
nasal & lacrimal glands:	vasoconstriction	abundant secretion
liver:	glucose production	---
kidney:	renin secretion	---
vessels within muscle:	dilation	---
cutaneous & visceral vessels:	constriction	---
piloerector muscles:	contraction	---
sweat glands (cholinergic):	secretion	---
adipose cells:	lipolysis	---
adrenal medulla:	epinephrine secretion	---

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

“**P**oint and **S**hoot”

Erection: **P**arasympathetic

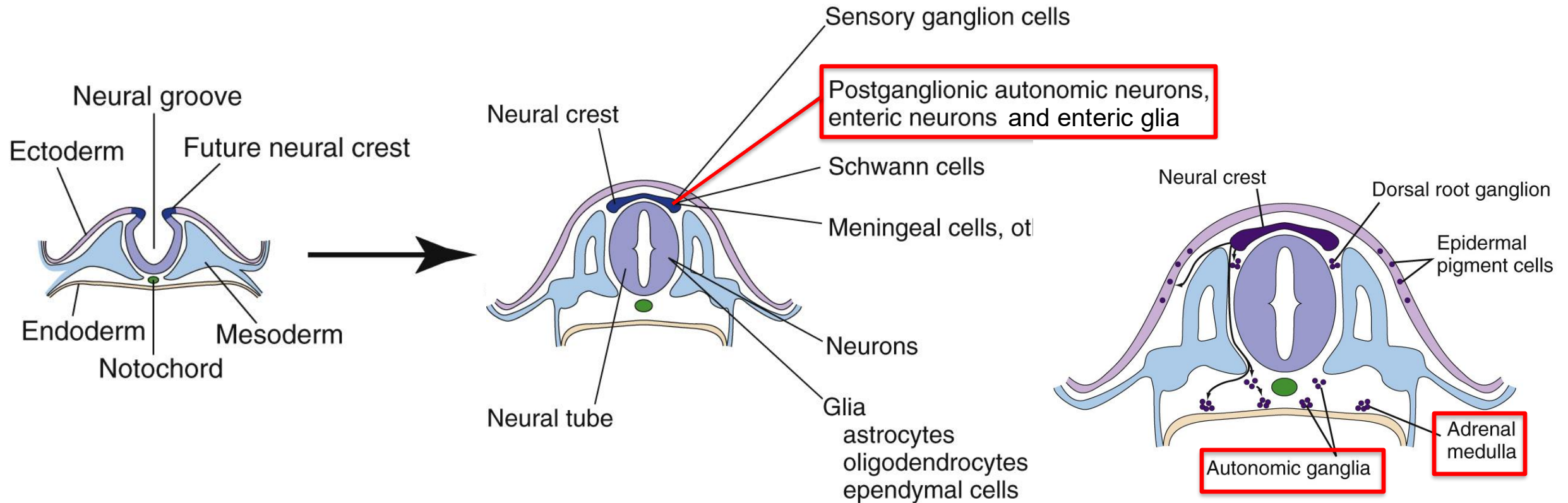
Ejaculation: **S**ympathetic

Dual innervation of many organs
— having a brake and an
accelerator provides more control

ANS development

NEW YORK INSTITUTE
OF TECHNOLOGY

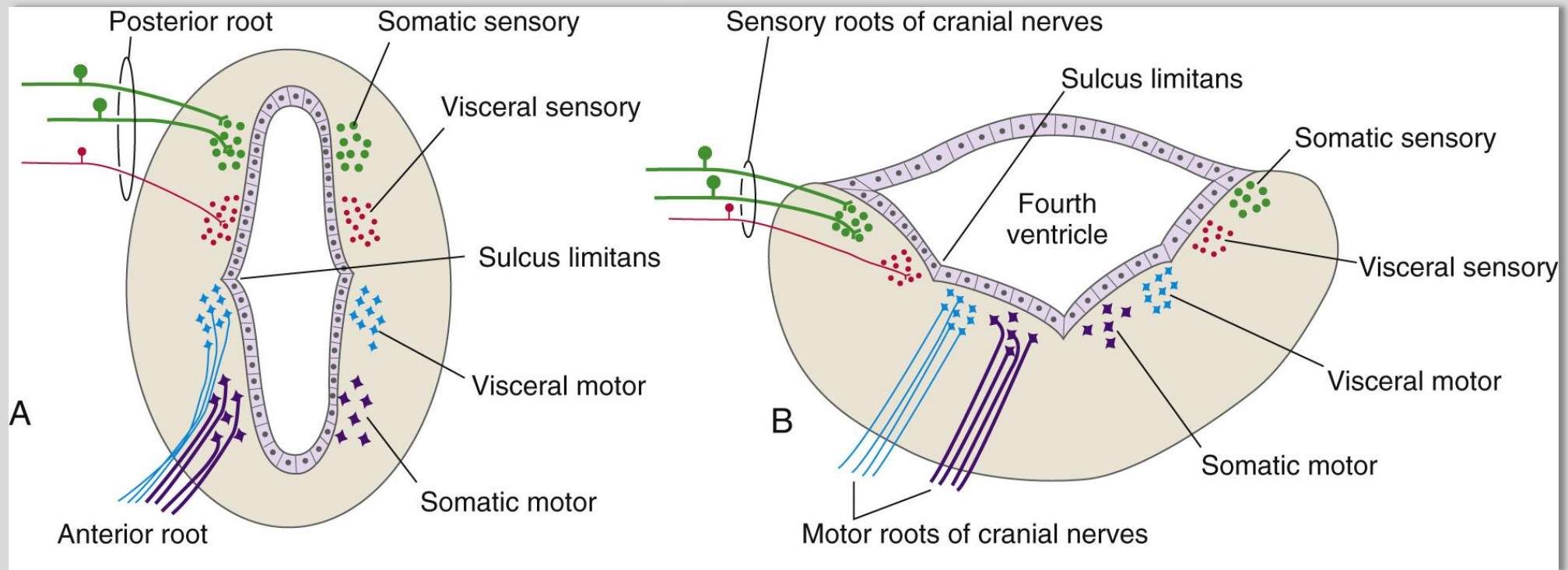
College of Osteopathic
Medicine

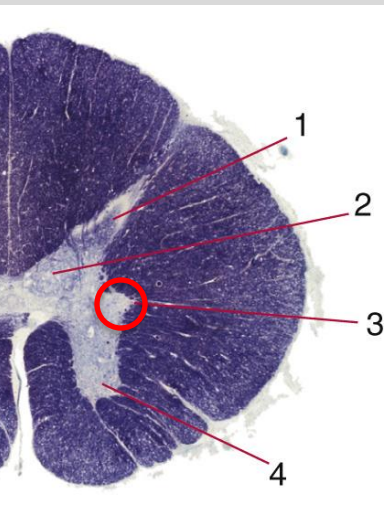


ANS development

**NEW YORK INSTITUTE
OF TECHNOLOGY**

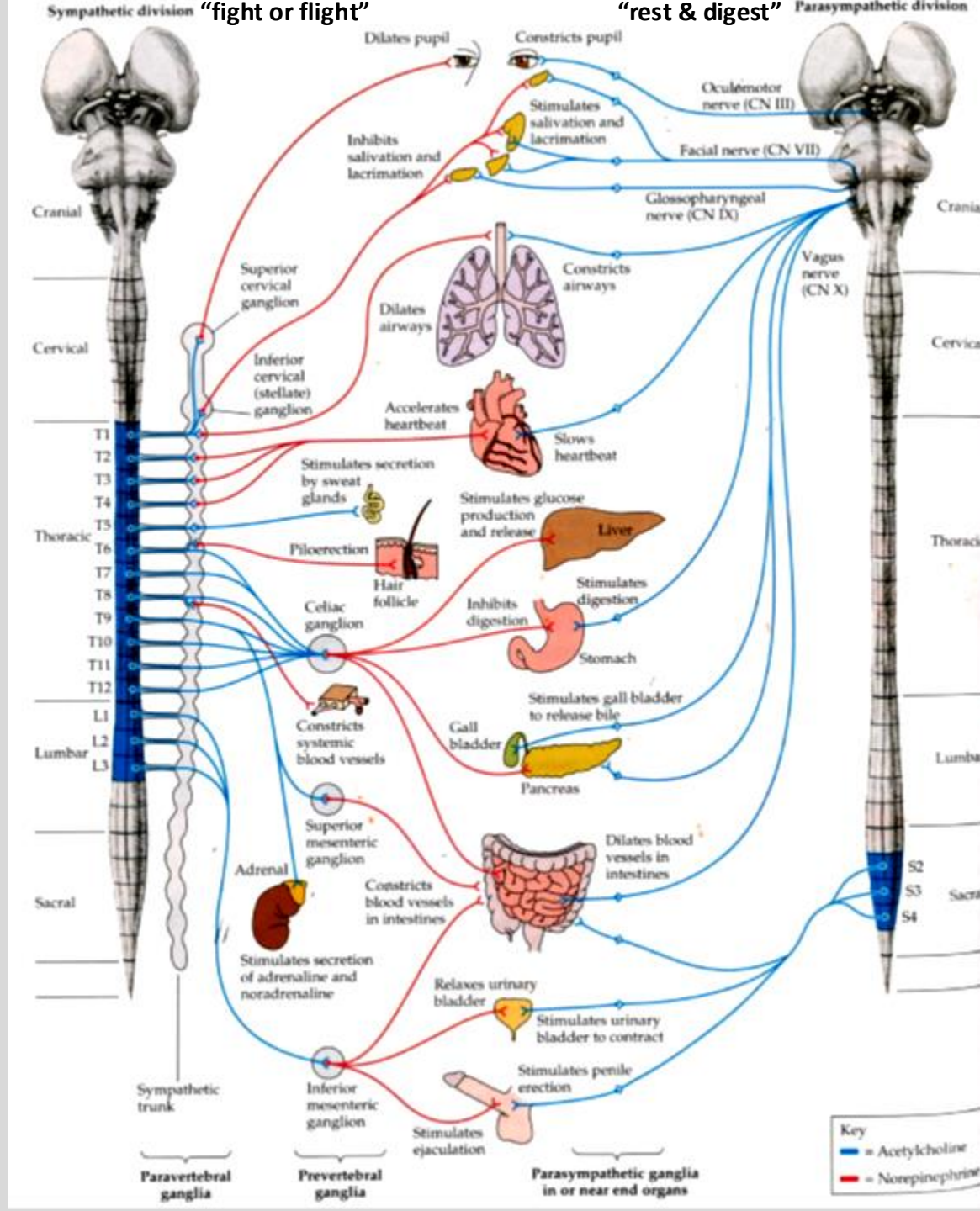
College of Osteopathic
Medicine





Intermediolateral cell column

T
o
r
a
c
o
l
u
m
b
a
r



C
r
a
n
i
o

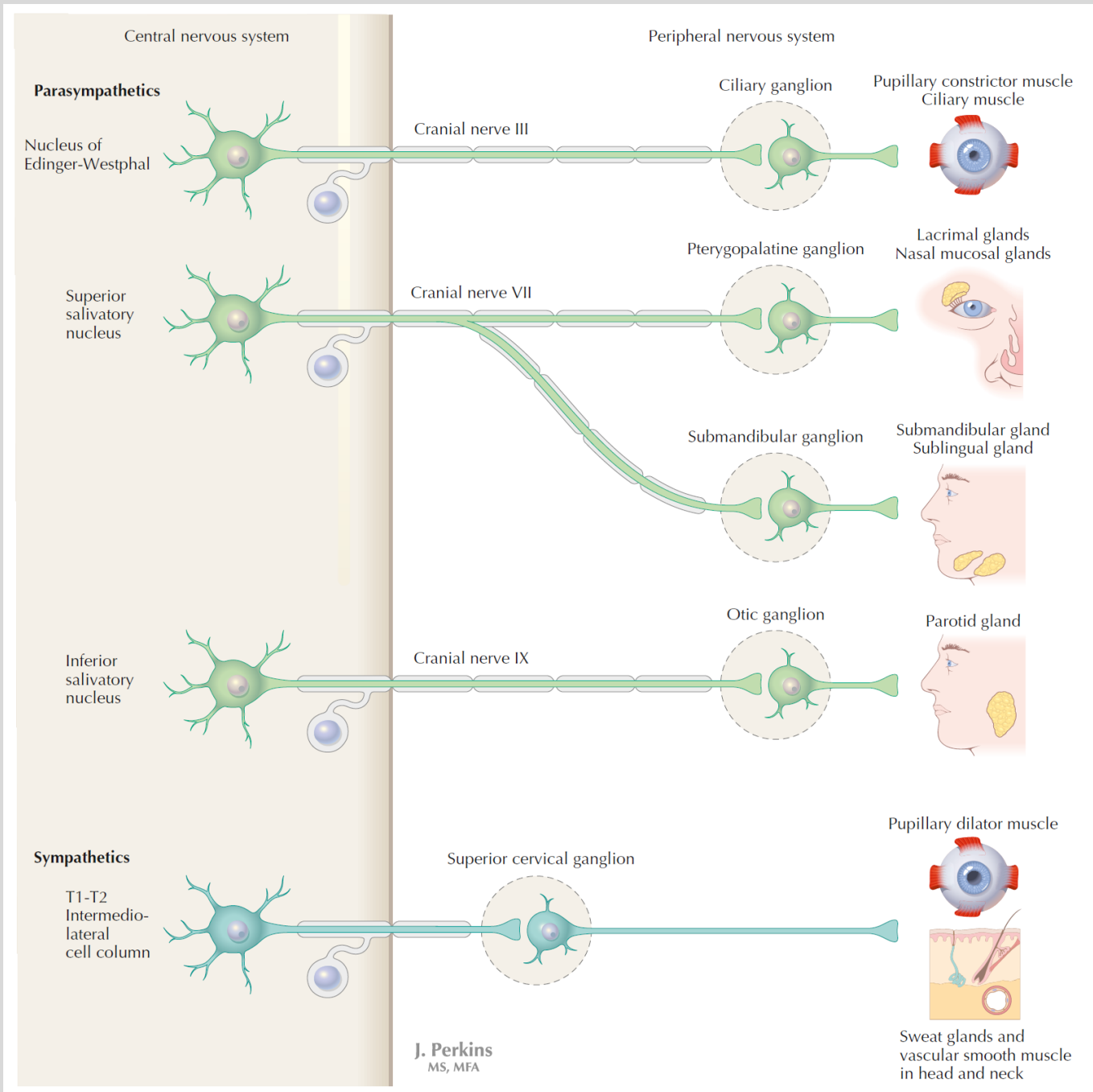


Dorsal motor nucleus of Vagus n.

Intermediolateral cell column

S
a
c
r
a
l





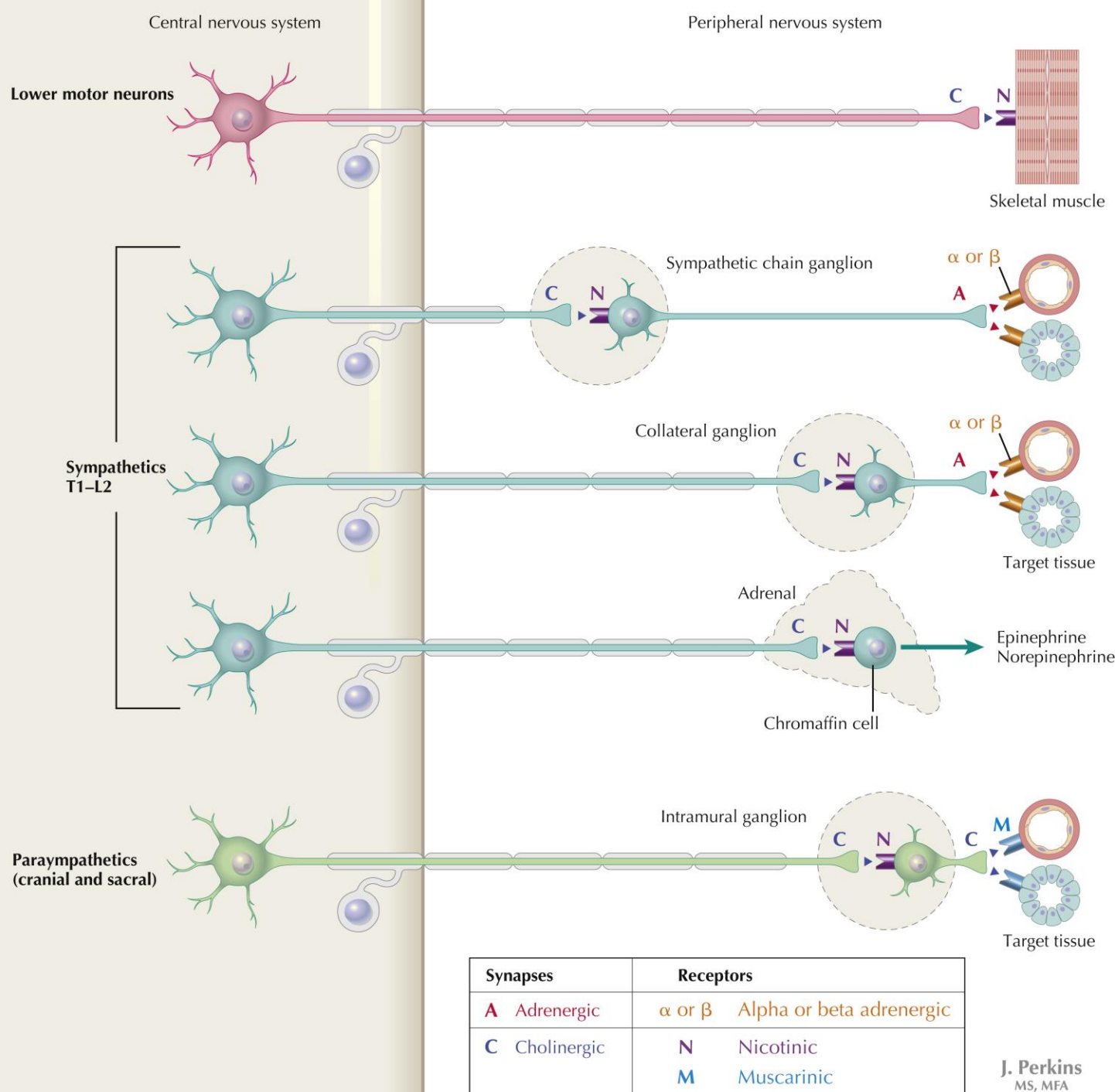
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine



Horner syndrome happens after sympathetic injury and it is the clinical triad of ipsilateral:

1. Ptosis (droopy eyelid),
2. Miosis (constricted pupil), and
3. facial anhidrosis (sometimes).

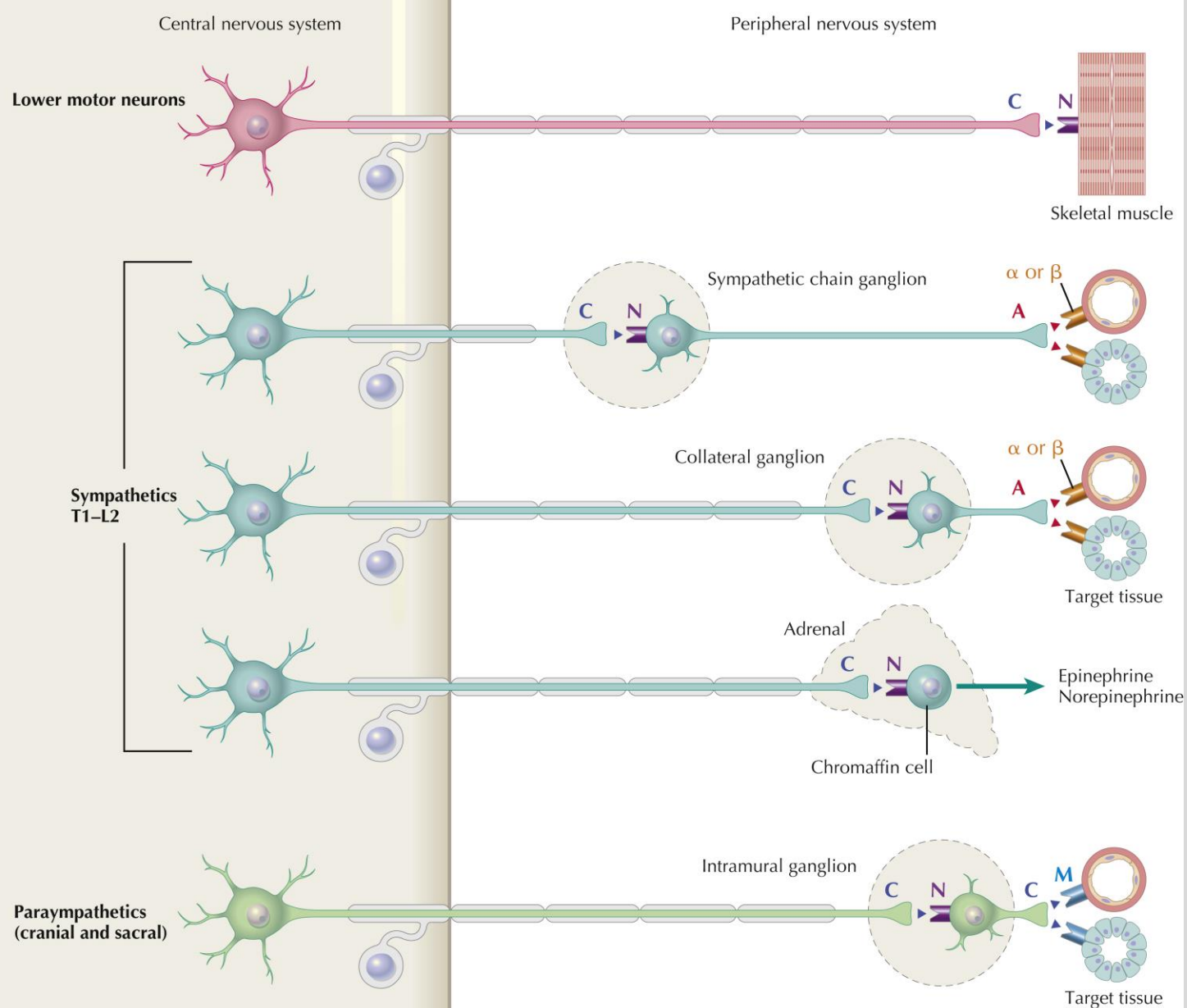


NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Similarities between Sympathetic & Parasympathetic

- Both are visceromotor and autonomic
- Both involve 2 neurons that synapse in a peripheral ganglion
- Both have myelinated preganglionic and unmyelinated postganglionic neuron
- Both have cholinergic (release ACh) preganglionic neuron
- Both have nicotinic receptors on the postganglionic neuron
- Both have excitatory preganglionic neuron



Synapses	Receptors
A Adrenergic	α or β Alpha or beta adrenergic
C Cholinergic	N Nicotinic M Muscarinic

J. Perkins
MS, MFA

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Differences between Sympathetic & Parasympathetic

- Preganglionic neuron body location in the CNS
 - Sympathetic is Thoraco-Lumbar
 - Parasympathetic is Cranio-Sacral
- Length of the pre- and post-ganglionic neurons:
 - Symp. has short pre- and long post-ganglionic
 - Parasymp. has long pre- and short post-ganglionic
- Postganglionic neurons:
 - Parasympathetic are cholinergic (release ACh)
 - Sympathetic are adrenergic (release NE) with **2 exceptions: sweat gland innervation** (release Ach) **and adrenal medulla** (release NE and Epi)
- The effect depends on the receptors expressed in the target tissue, i.e. could be excitatory or inhibitory

The effect depends on the receptors

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Autonomic Metabotropic Receptors

Muscarinic receptors:

Parasympathetic

Receptor	G-protein	Messenger	Effect	Example Functions
M ₁	G _q	Phospholipase C & IP ₃ increased	elevated Ca ⁺⁺	salivary gland & stomach secretion
M ₂	G _i	cAMP decreased	decreased Ca ⁺⁺ influx & increased K ⁺ efflux	decreased heart rate and force
M ₃	G _q	Phospholipase C & IP ₃ increased	elevated Ca ⁺⁺	constriction of vessels & bronchioles detrusor contraction (micturition)
M ₄	G _i	cAMP decreased	decreased Ca ⁺⁺ influx & increased K ⁺ efflux	inhibitory effects
M ₅	G _q	Phospholipase C & IP ₃ increased	elevated Ca ⁺⁺	present in CNS

Sweat gland secretion (Ach released by sympathetic postganglionic neurons)

Adrenergic receptors:

Sympathetic

Receptor	G-protein	Messenger	Effect	Example Functions
α ₁	G _q	Phospholipase C & IP ₃ increased	elevated Ca ⁺⁺	cutaneous & GI vasoconstriction; urethral sphincter contraction
α ₂	G _i	cAMP decreased	decreased Ca ⁺⁺ influx	inhibition of neurotransmitter release
β ₁	G _s	cAMP increased	elevated Ca ⁺⁺	increased cardiac output (rate & force)
β ₂	G _s G _i	cAMP increased cAMP decreased	elevated Ca ⁺⁺ decreased Ca ⁺⁺ influx & increased K ⁺ efflux	constriction of gut sphincters muscle vessel dilation; detrusor relaxation
β ₃	G _s G _i	cAMP increased cAMP decreased	elevated Ca ⁺⁺ decreased Ca ⁺⁺ influx & increased K ⁺ efflux	adipose tissue lipolysis detrusor relaxation

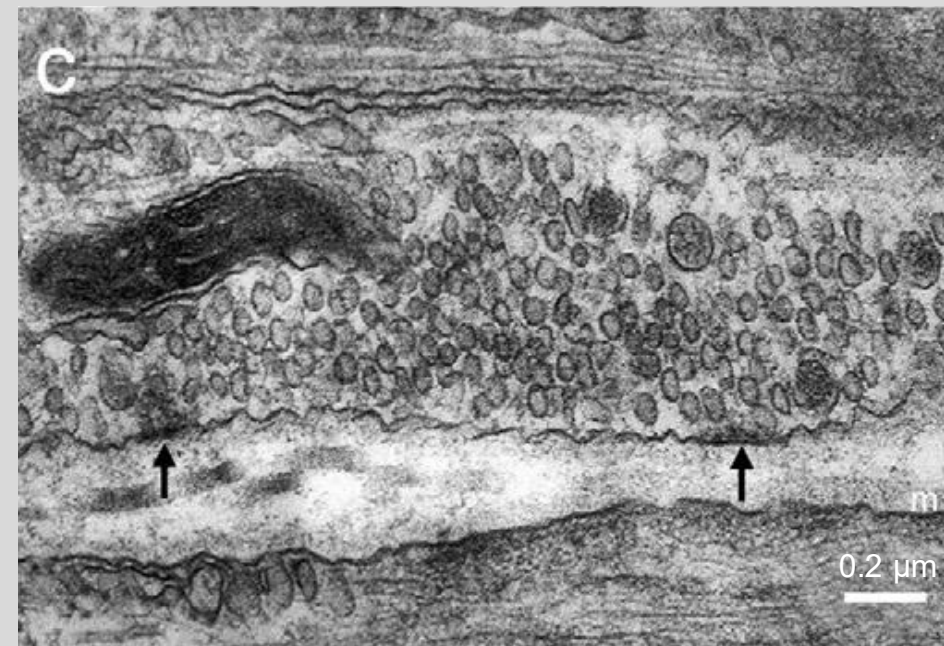
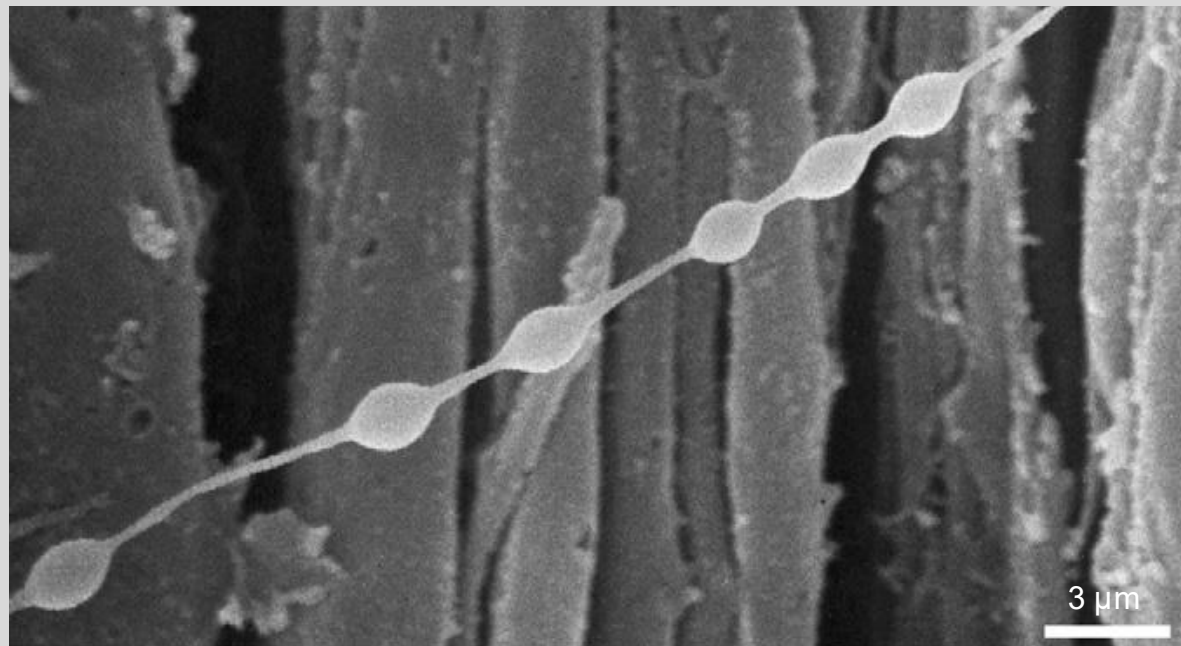
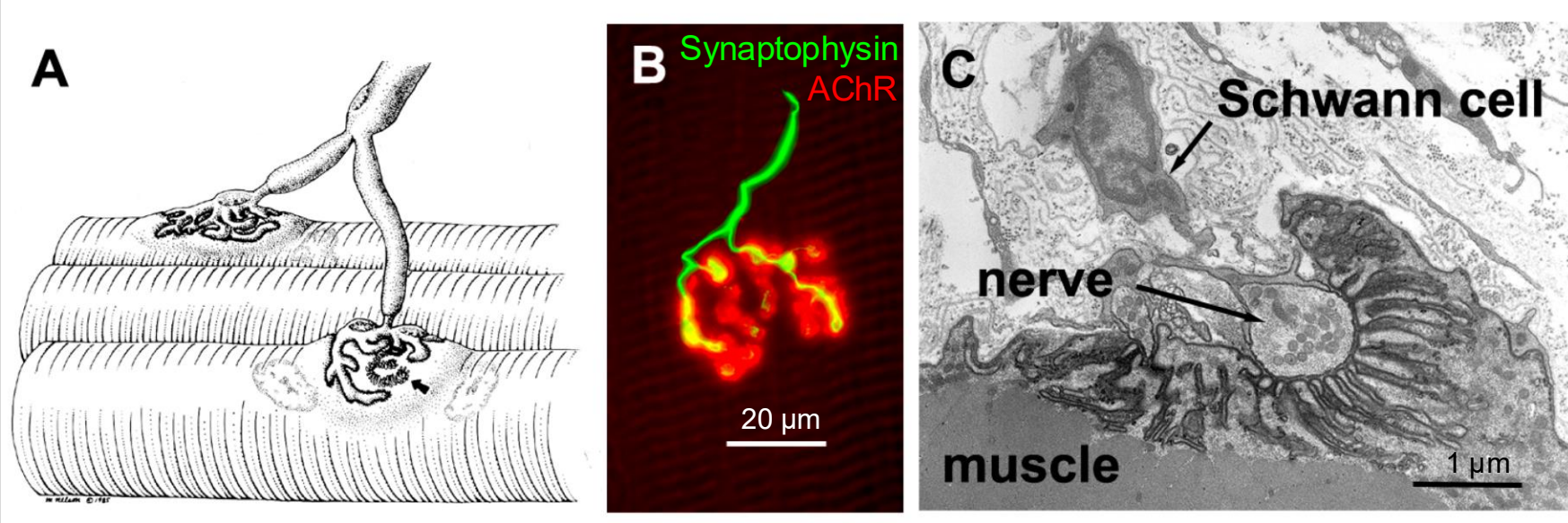
<http://vanat.cvm.umn.edu/ans/pages/Physiol.html>

Somatic vs Autonomic Muscle Innervation

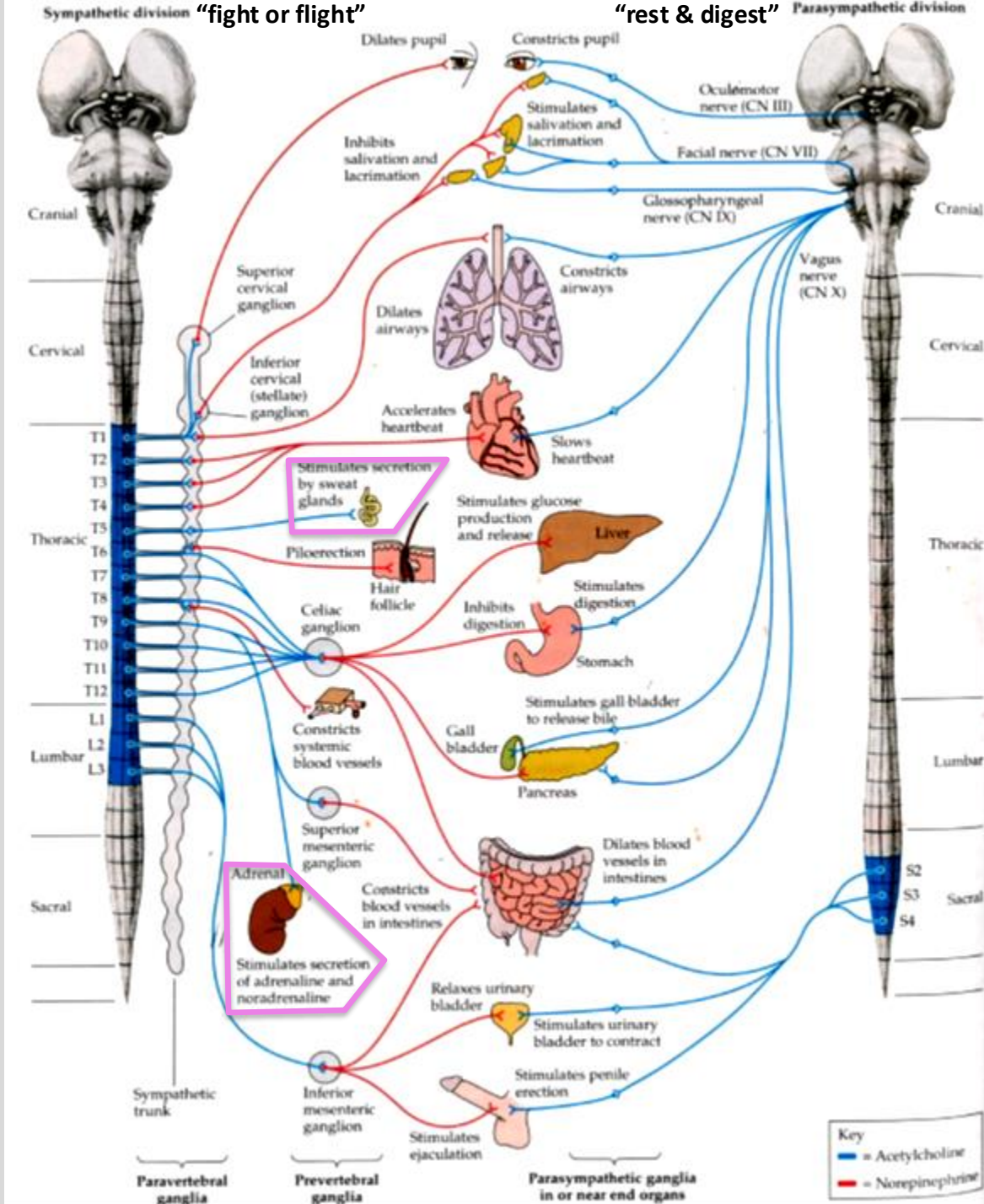
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Int J Mol Sci. 2017; 18(10): 2183.



Neurochem Int. 2008;
52(1-2): 14-25.



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Sweat glands secretions activated by sympathetic postganglionic cholinergic neurons (via M₃).

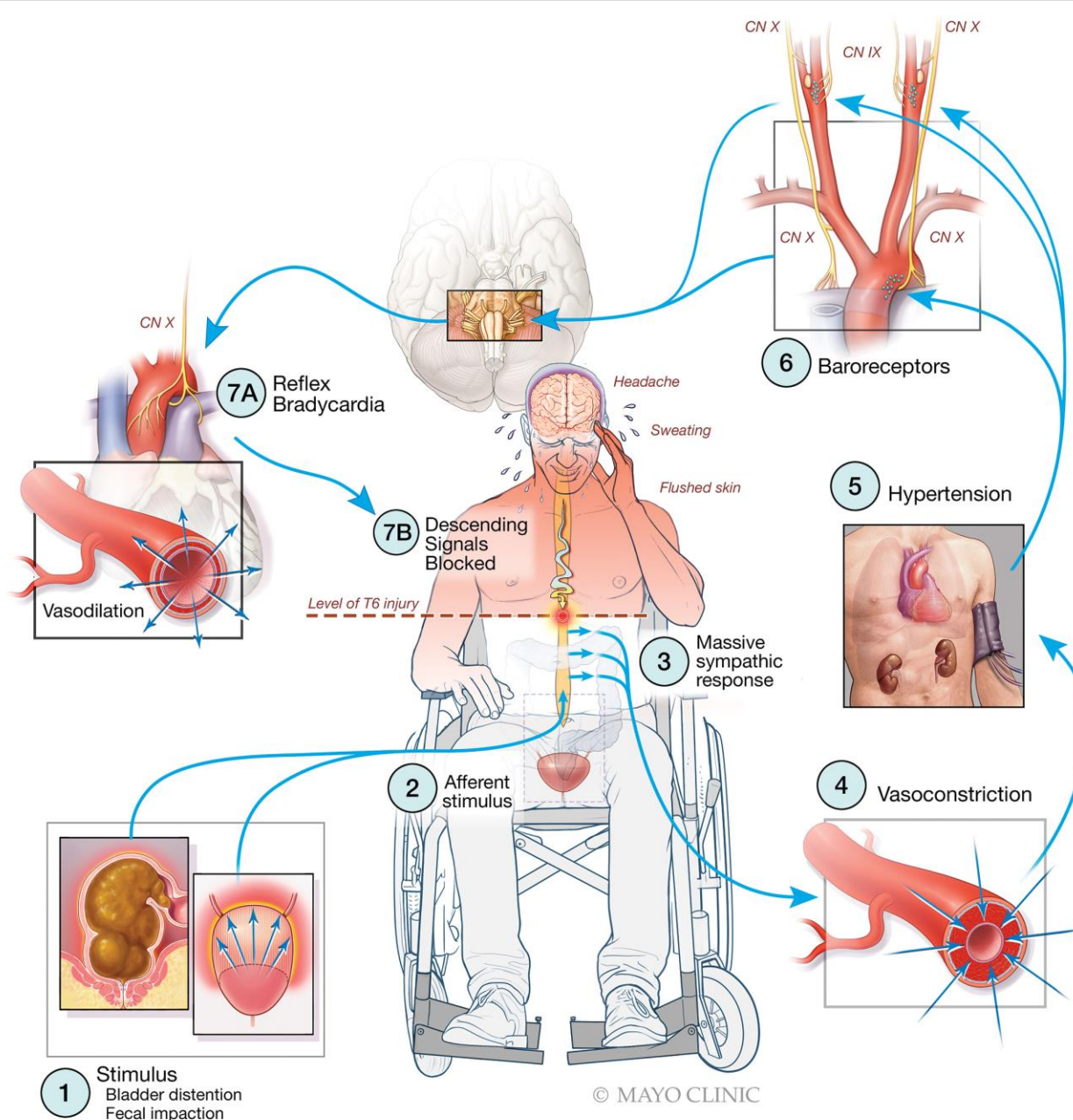
The adrenal medulla:

- Chromaffin cells – neuroendocrine cells
- Secretes epinephrine (80%) and norepinephrine into the bloodstream
- Sympathetic effect sustained

Autonomic Dysreflexia

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

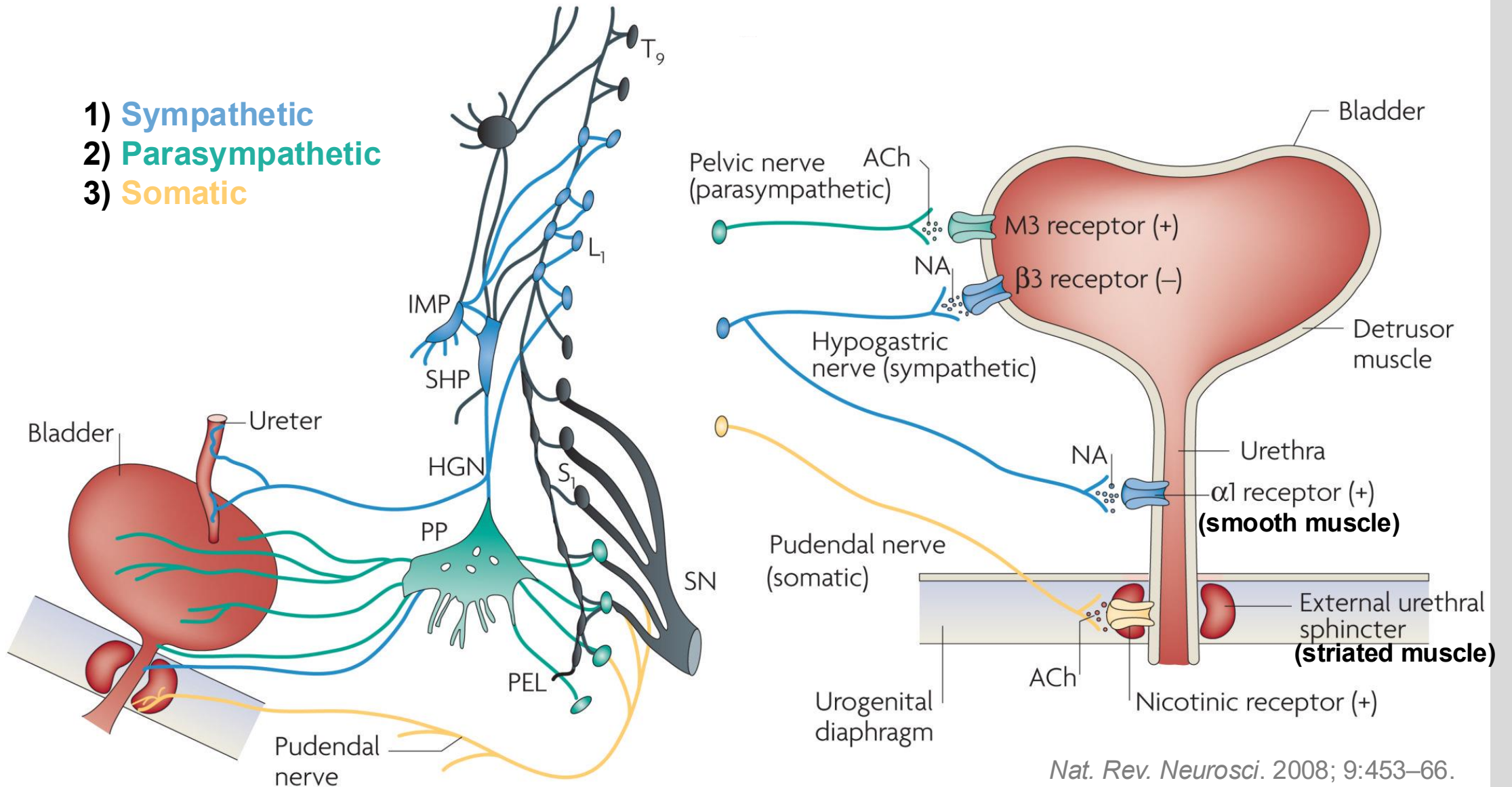


- Consequent to **spinal cord injury**, typically **above T6**
- Hyper-sensitivity of autonomic reflexes to sensory stimuli (i.e. **autonomic spasticity**)
- Sympathetic surge causing widespread vasoconstriction → peripheral arterial hypertension → detected by baroreceptors in the neck → parasympathetic slow heart rate
- **Uncontrolled hypertension and bradycardia!**
- Life-threatening

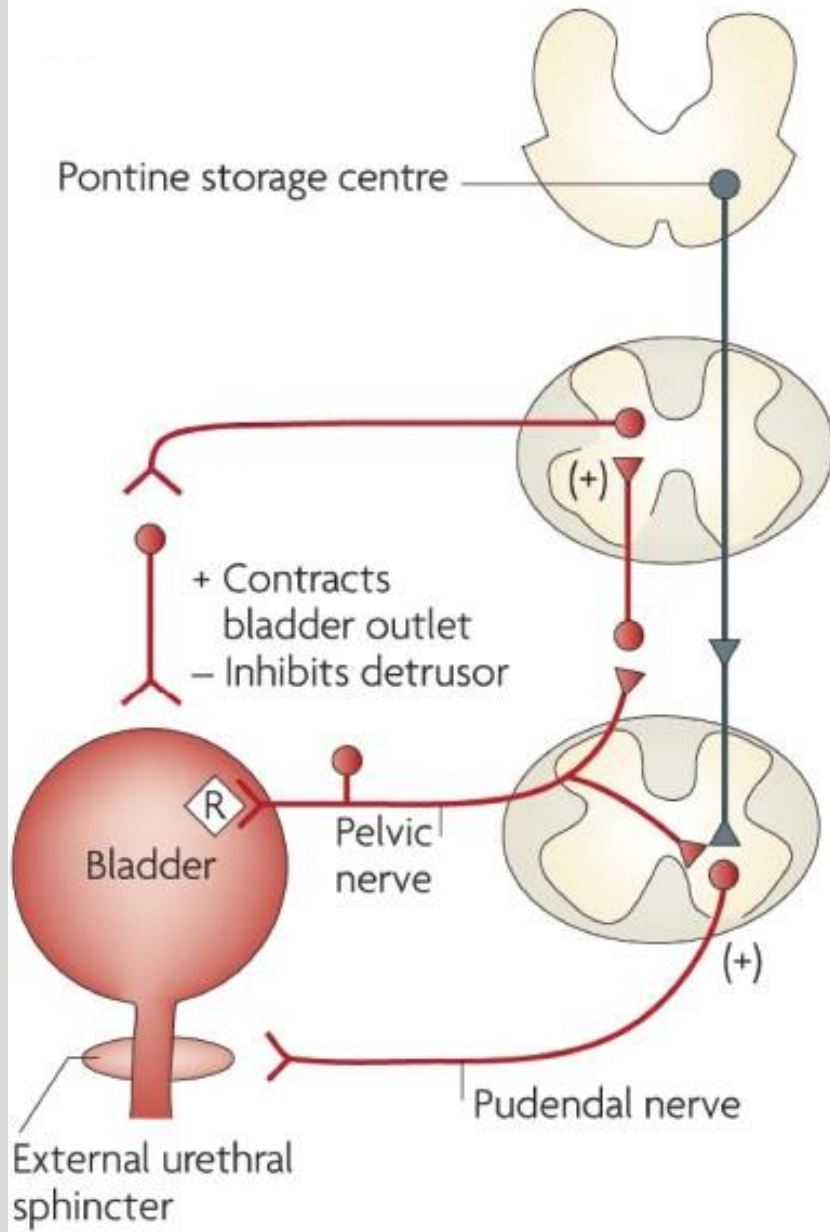
Efferent innervation of the lower urinary tract

NEW YORK INSTITUTE
OF TECHNOLOGY

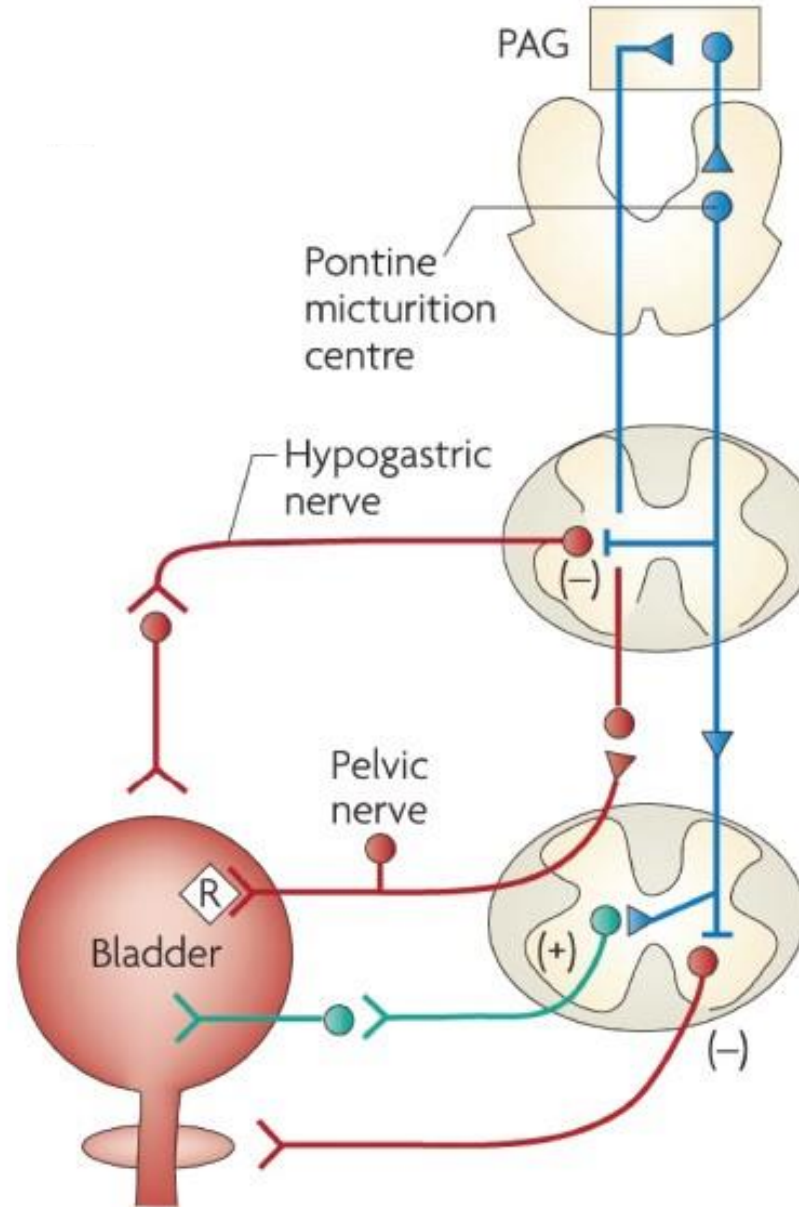
- 1) Sympathetic
- 2) Parasympathetic
- 3) Somatic



Urine storage reflexes



Voiding reflexes

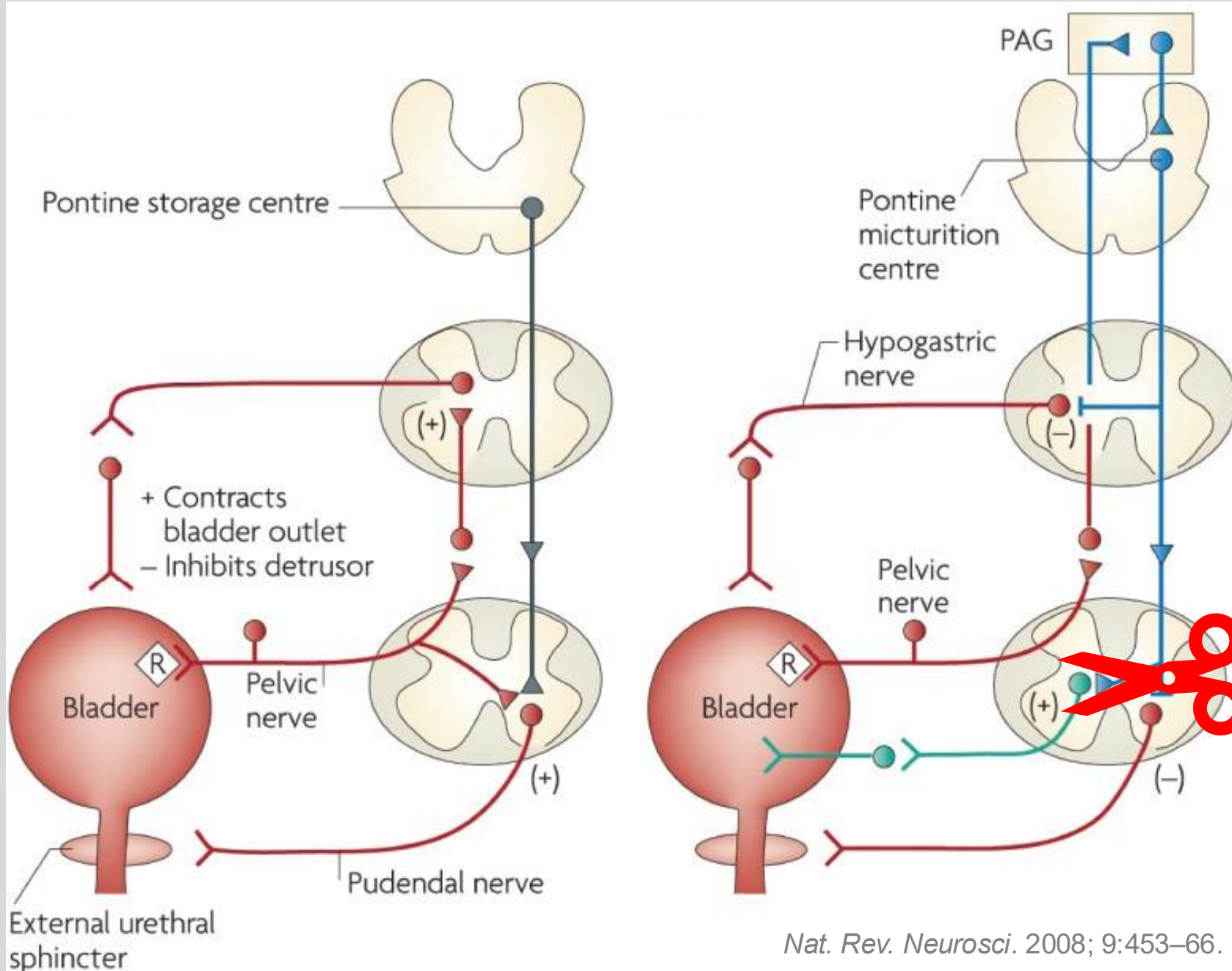


Nat. Rev. Neurosci. 2008; 9:453–66.

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Atonic bladder / Lower Motor Neuron bladder



Nat. Rev. Neurosci. 2008; 9:453–66.

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Lesions of the **conus medullaris**, **cauda equina** or **sacral cord** are a lot like LMN damage:

Detrusor is unable to contract → bladder expands under relatively low pressure → bladder fills up but never completely empties

External sphincters not contracted → small amount of urine may slowly leak out

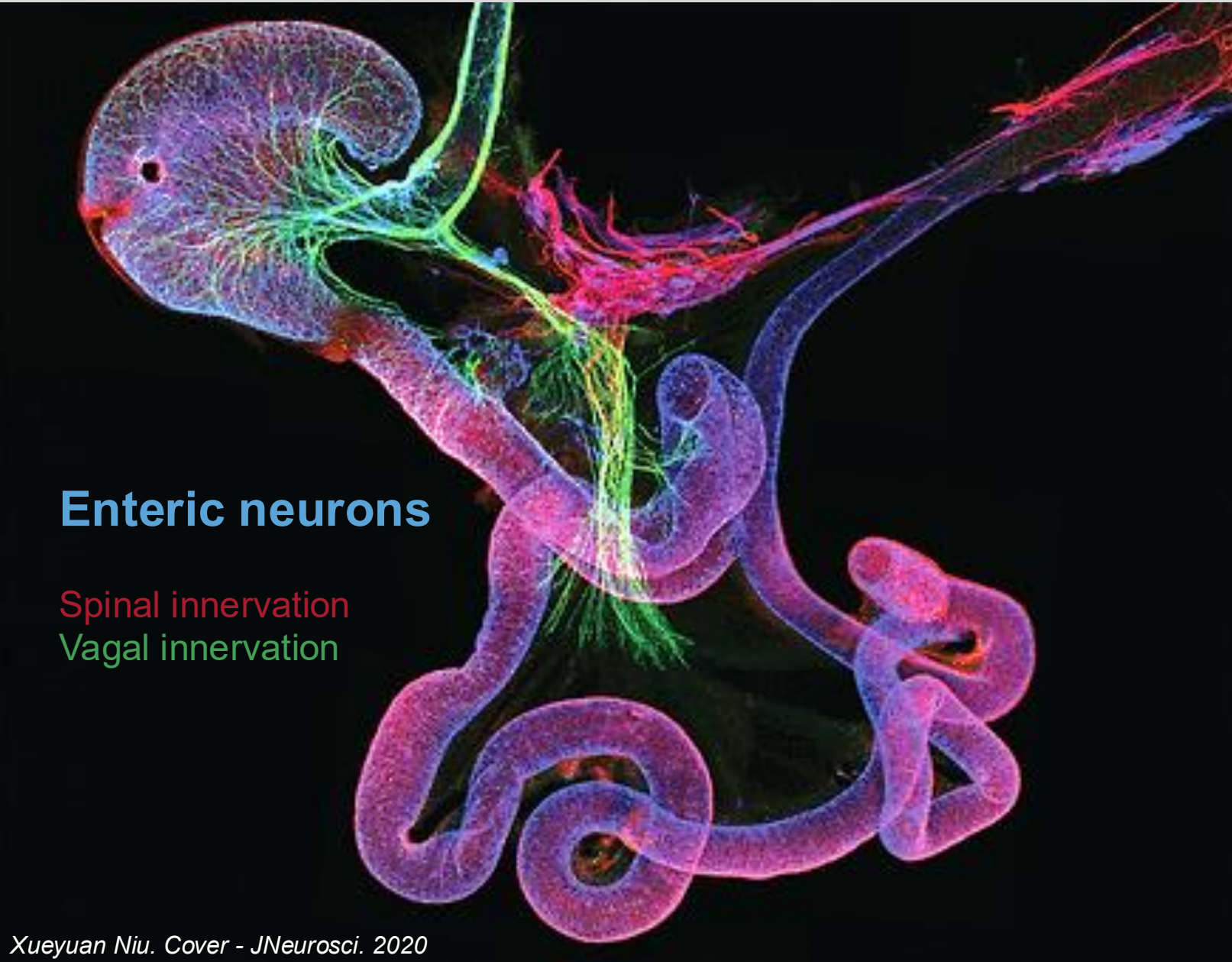
The enteric nervous system (ENS)

**NEW YORK INSTITUTE
OF TECHNOLOGY**

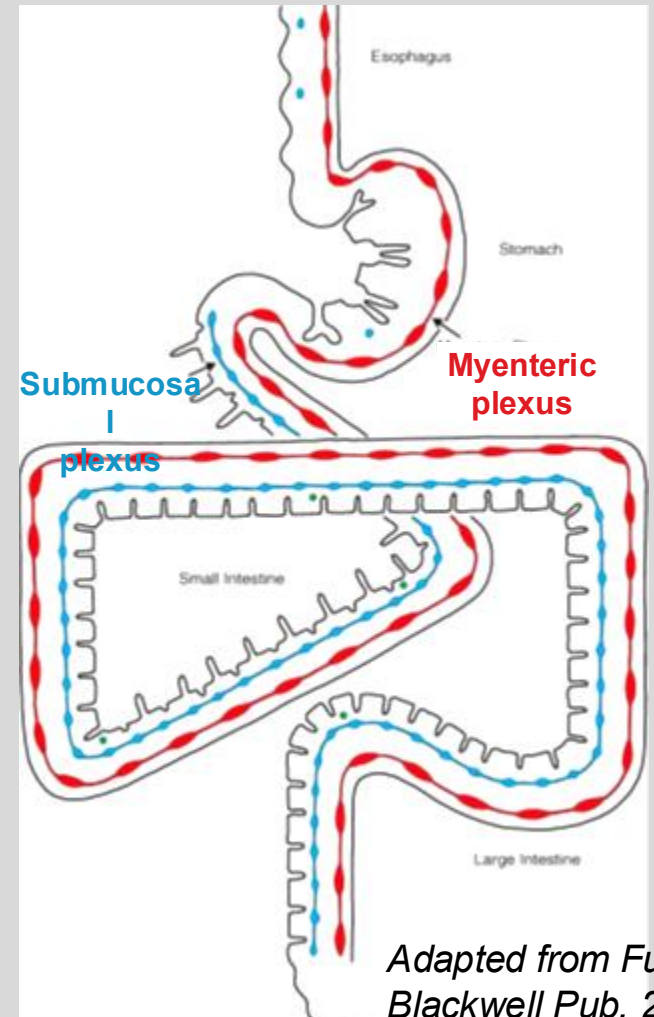
College of Osteopathic
Medicine

Enteric neurons

Spinal innervation
Vagal innervation



Xueyuan Niu. Cover - JNeurosci. 2020

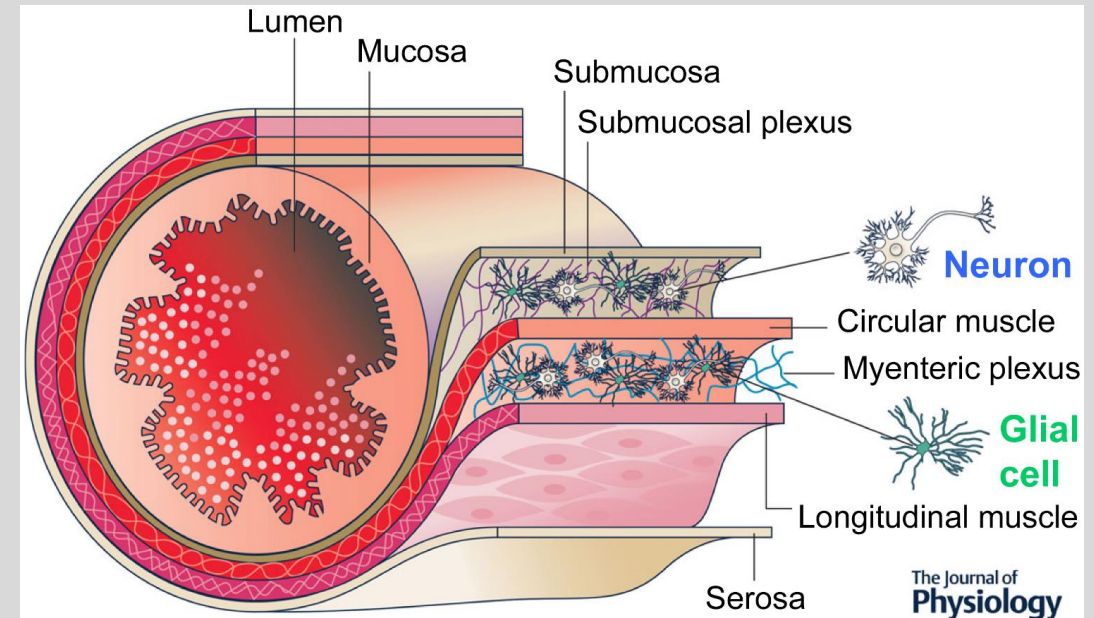
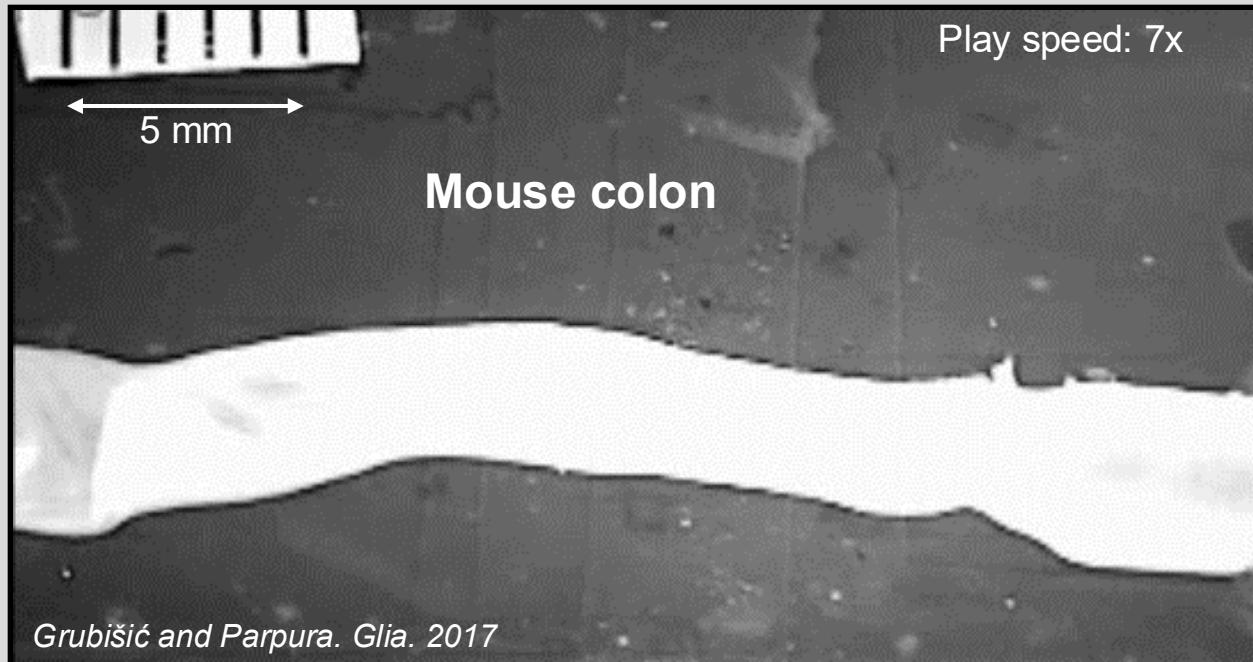


Adapted from Furness.
Blackwell Pub. 2006.

ENS – 3rd division of the ANS, “the brain in the gut”

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



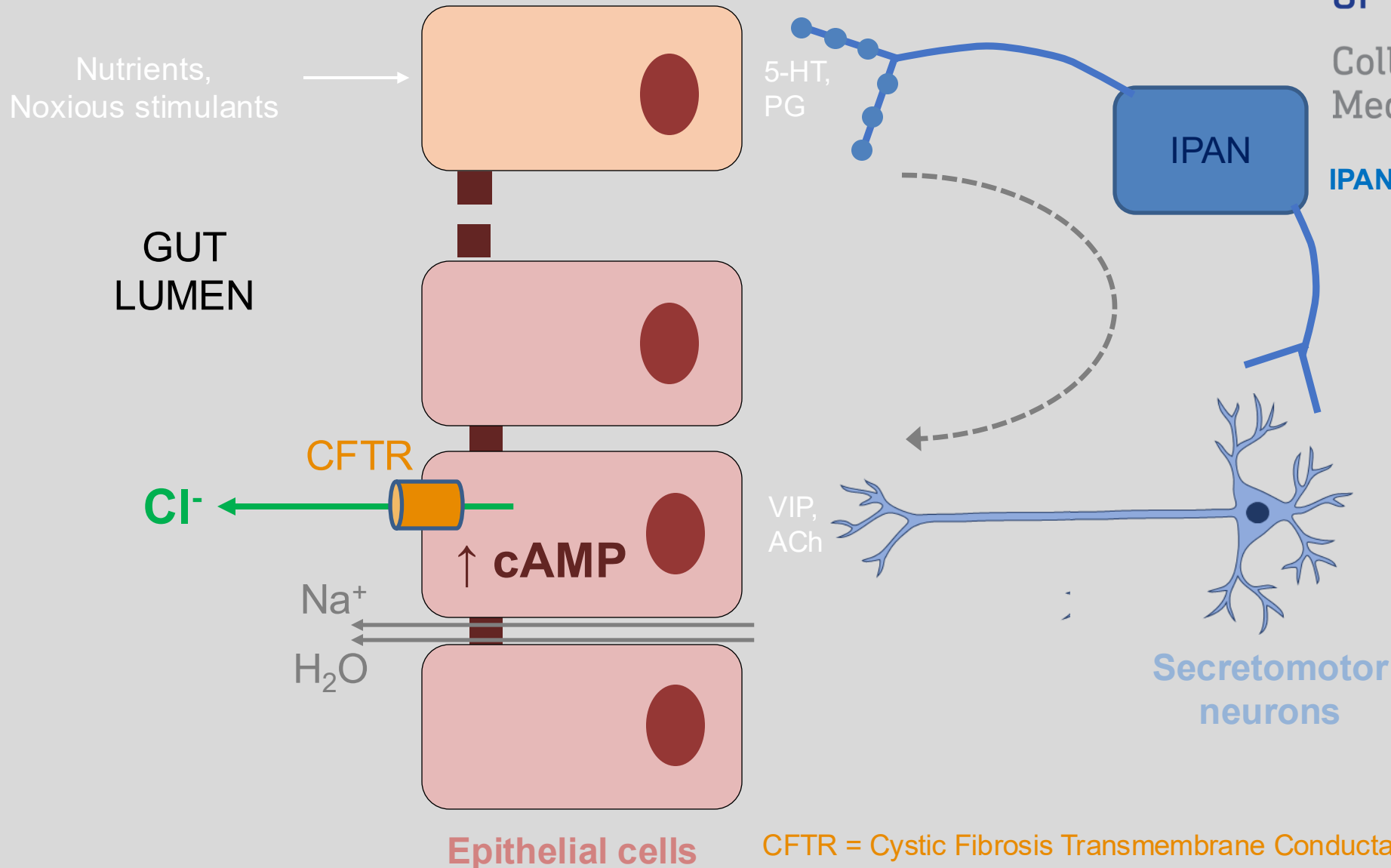
Grubišić and Gulbransen. J. Physiol. 2017.

Secretomotor reflex

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

IPAN = Intrinsic Primary Afferent Neuron



Summary

1. ANS and Homeostasis
2. “Rest and digest” (Parasympathetic) vs “Fight or Flight” (Sympathetic)
3. ANS centers in the brain stem and hypothalamus
4. Similarities and differences between the Sympathetic and Parasympathetic divisions
5. Actions of the muscarinic and adrenergic receptors
6. Somatic vs autonomic muscle innervation
7. Baroreceptor reflex
8. Orthostatic hypotension
9. Carotid Sinus Massage
10. Autonomic Dysreflexia
11. Innervation and reflexes of the urinary system
12. Spastic and atonic bladder
13. ENS networks: Myenteric (Auerbach’s) and Submucosal (Meissner’s)
14. Gut motility and secretomotor reflexes
15. Hirschsprung's Disease